PROJECT PROPOSAL

BLOCK 1D-2024-2025

SmartPredict:

Financial Forecasting for Competitive Trading

Name of the Group

Group 17

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I. Introduction

Our project seeks to address an issue faced by Move Tickers by leveraging machine learning, which we believe is the most effective solution. No other field can provide the unique method of leveraging data to accomplish this as machine learning does. It enables tasks that would take a great deal of time to do using pen and paper to be completed in a matter of seconds. These factors enable the project to provide distinctive and worthwhile insights on the financial industry, thus resolving Move Ticker's issue.

II. Problem Statement

The stock market is very risky to enter. You can make quite a bit of money, but simultaneously you run the risk of losing it all. People often go to specialized companies for help with the stock market. And to attract customers, companies need to make their customers money. So how can we improve the chances for the customers of Move Tickers to make money while trading on the stock market? That's where our team comes in. Our goal is to improve the chances of Move Ticker's customers through machine learning. We will create business value through the insights of our model, which can reduce the time needed for people to research the financial records of companies they plan on investing in. Our objective is to have a prediction of the price which can be easily interpreted and understood by a person with basic financial knowledge. We can achieve our goals through creating an AI model to predict the price, which will require modeling and data cleaning/formatting, and creating a dashboard which gives all the required information, for which UI/UX design will be needed. The outcome of the model should be a simple prediction, and of the dashboard to have it be understandable. This will give Move Tickers the competitive edge they need in the market of financial companies.

III. Data Description

In our project, we will use six datasets that help us understand the stock market from different perspectives. These datasets come from public sources like Yahoo Finance, the World Bank, the European Central Bank, and Kaggle. We will use **NASDAQ-100 daily and minute-level stock data**, which includes key features like open, high, low, close prices, volume, and timestamps. The daily data shows long-term trends, while the minute-level data captures short-term price movements. To provide a broader market context, we are including **NASDAQ-100 index data**. Since many NASDAQ-100 companies operate internationally, we will also use **USD exchange rate data** (against major currencies like EUR and JPY) to observe global financial pressures. To capture economic conditions, we are using **macroeconomic indicators** such as interest rates, inflation (CPI), and the VIX index. These indicators reflect investor sentiment and market risk.

A key feature in our model will be EPS *(Earnings Per Share) data*, which includes actual earnings, analyst estimates, and surprise values. We will merge these with stock data to study how financial results affect market behavior. All datasets are **time-series** and will be joined using the **date** field. The data is structured in CSV format and organized by time rows with multiple features per column. During preprocessing, we will handle missing values using forward-fill or interpolation depending on the data frequency. We will also standardize all numeric values for consistency. We expect potential issues like missing values, misaligned timestamps, and outliers. These will be fixed during preprocessing. Feature engineering will include daily returns, moving averages, price volatility, and EPS surprise.

Our goal is to create a clean and consistent dataset that supports both prediction and explanation in financial decision-making.

The following table summarizes the datasets we plan to use, their sources, structure, and how contribute to our project:

Dataset Name	***Source***	***Key Features***	***Frequency***	***Use Case***
NASDAQ-100 Daily Data	Yahoo Finance	Open, High, Low, Close, Volume	Daily	Long-term trend analysis
NASDAQ-100 Minute Data	Yahoo Finance	OHLC, Volume, Timestamp	1-minute intervals	Intra-day behavior, short-term market reactions
NASDAQ-100 Index Data	Yahoo Finance	Index value, Volume	Daily	Market-wide momentum indicator
Exchange Rates (USD/FX)	ECB / Yahoo Finance	USD vs EUR, JPY, GBP, etc.	Daily	Global financial signal for international companies
Macroeconomic Indicators	World Bank, FRED	Interest rate, Inflation (CPI), VIX	Monthly/Daily	Market sentiment, investor behavior
EPS Reports	Kaggle	EPS Actual, Estimate, Surprise	Quarterly	Event-based predictors for price movement

IV. Methodology

Data Collection & Pre-processing:

1. **Data Sources:**
   - NASDAQ-100 Daily and Minute Data (10 years): Used to capture stock-level behavior at different time scales.
   - NASDAQ-100 Index Data: Provides market context and macro-level momentum.
   - Exchange Rates: Useful for international companies in the NASDAQ-100, where currency fluctuations may impact earnings.
   - Macroeconomic Indicators: Signals like interest rates, inflation, and VIX help explain investor sentiment and market risk.
   - Yahoo Finance EPS Data: We merge quarterly earnings announcements (EPS Actual, Estimate, and Surprise) with stock price data to build event-based predictors.
2. **Pre-processing Steps:**
   As a first step, we will join multiple datasets from different sources — NASDAQ stock prices, earnings reports, macroeconomic indicators, exchange rates, and global stock indices — using the date as the primary key. This ensures all data sources are synchronized on a common daily timeline, enabling time-series modeling.

   After merging the datasets, we will conduct a correlation analysis to explore relationships between key variables, such as how macroeconomic trends or EPS results influence stock price movement. This will help in identifying and selecting the most helpful features for modeling.

   We will then perform feature engineering to extract meaningful predictors. One key feature is EPS Surprise, calculated as the difference between the reported EPS and the analyst estimate. Additional features will include technical indicators (e.g., daily returns, moving averages, price volatility), macroeconomic trend shifts (e.g., changes in interest rates or inflation).

   To handle missing values, we will apply forward-fill or interpolation techniques depending on the data frequency and nature. Lastly, all numerical features will be standardized to ensure they are on a consistent scale, which is critical for optimizing machine learning model performance.

Model Selection & Training:

Our project aims to address two closely related predictive tasks:

1. **Classification** – to determine whether a stock's price will increase or decrease on the day following an earnings announcement.
2. **Regression** – to estimate the exact closing prices of the stock in the future

To tackle the **classification** task, we will begin with a Logistic Regression model as a baseline due to its simplicity and interpretability. After that we will explore Random Forest and XGBoost, which are well-suited for structured financial data and capable of capturing complex interactions among features. These models also provide useful insights into feature importance, which can inform both prediction and investment strategies.

For the **regression** task, we will use models such as Linear Regression as a baseline, followed by Gradient Boosting Regressors and potentially a feedforward neural network. These methods will help us estimate the magnitude of the price change, not just its direction.

The dataset will be split into the **training and testing set,** using 80% of the earliest data for training and the remaining 20% for testing. To ensure robustness, we will apply 5-fold cross-validation within the training set. Additionally, we will use time-aware validation, where models are trained on past data and tested on future data, to reflect real-world application scenarios.

By using both classification and regression, we can build a more complete prediction system. It will not only tell us whether a stock's price is likely to go up or down but also give an estimate of what the price will be. This helps traders and analysts make better decisions.

Evaluation Metrics

To assess the performance of our models, we will use different evaluation metrics for the classification and regression tasks.

For the **classification** task, we will use accuracy to measure how often the model predicts the correct direction (up or down). We will also use precision, recall, and F1 score to better understand performance, especially if the data is imbalanced.

For the **regression task**, we will evaluate the models using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE gives us the average size of the errors in our predictions, while RMSE penalizes larger errors more heavily, helping us detect when the model is making particularly bad predictions.

Together, these metrics will give us a detailed understanding of how well our models are performing and help us compare different approaches effectively.

Deployment

Firstly, we will present the results of our models through an interactive dashboard. This dashboard will display key metrics such as predicted price movements, estimated future

prices, confidence scores, and important features influencing the predictions. It will allow users, such as analysts and traders, to explore the model's outputs in a clear and actionable format.

If the model performs well and proves reliable over time, we plan to develop a chatbot-based assistant. This bot will use the model's predictions to provide daily trading recommendations, such as whether to buy, or sell specific stocks based on upcoming earnings reports and market conditions. The bot will be designed to integrate into Move Tickers' existing platform, providing real-time insights and helping users make informed trading decisions.

V. Project Timeline

Phase 1: Data Collection & Preprocessing
Duration: Week 1 (Mon-Sat)
Milestone: Final dataset ready for modeling
Deliverables: Cleaned, engineered and merged dataset in .csv format
Week 4: Monday (Data Engineer, Data Scientist)

- Datasets:
  - NASDAQ-100 Minute Data
  - NASDAQ-100 Daily Data
  - NASDAQ-100 Index Data
  - Exchange Rate Trends
  - Macroeconomic Indicators
  - Dataset for historical EPS of NASDAQ stocks (source: Kaggle)
- Data cleaning:
  - Look for missing data
  - Replace missing values

Week 4: Tuesday / Wednesday (Data Engineer, Data Scientist)

- Feature Engineering:
  - Daily returns
  - Moving averages
  - Price volatility
  - Macroeconomic trends

Week 4: Thursday (Data Scientist)

- Preprocessing
  - Normalize numerical data
  - Merge datasets into one, align timestamps

Week 4: Friday (All members)

- Review
  - Review the dataset
  - Check for potential issues
  - Fix issues
- Export
  - Export final dataset into csv

Phase 2: Model Training & Evaluation

Duration: Week 2 (Mon–Sun)
Milestone: Best-performing models selected and evaluated
Deliverables: Trained ML models + evaluation metrics
Week 5: Monday/Tuesday (ML Engineer)

- Baselines
    - Train Logistic Regression as a baseline for classification
    - Train Linear Regression as a baseline for regression
- Model training
    - Train XGBoost, Random Forest for classification
    - Train Gradient Boosting Regressor, Feed Forward Neural Networks
    - Select best performing models for each task

Week 5: Wednesday (ML Engineer)

- Hyper tuning
    - Hyper-tune best performing models for each task
- Evaluation
    - Evaluate classification model using recall, precision, F1 score, confusion matrix
    - Evaluate regression model using MAE, RMSE

Week 5: Thursday (ML Engineer)

- Optional: re-train models
    - Improve models with gathered insights
    - Re-train
    - Re-evaluate

Week 5: Friday (ML Engineer)

- Optional: re-train models
    - Improve models with gathered insights
    - Re-train
    - Re-evaluate

Phase 3: MVP Development
Duration: Week 3 (Mon–Sun)
Milestone: Interactive MVP with integrated model logic
Deliverable: Streamlit or Gradio-based MVP + initial documentation
Week 6: Tuesday (UI/UX Designer, Frontend Dev, ML Engineer)

- UI/UX design
    - Create simple wireframe
- Initial dashboard
    - Create initial dashboard using Streamlit
    - Implement core ML logic

Week 6: Wednesday (Frontend Dev, ML Engineer)

- Integration
    - Integrate ML models with frontend
    - Develop basic input/output fields
    - Basic user input/output integration

Week 6: Thursday (DevOps, ML Engineer)

- Unit Testing
    - Set up Poetry for environment and testing
    - Write initial unit tests for ML pipeline and frontend routes
    - Debug and fix any integration issues

Week 6: Friday (Based on issue)

- Debug and fix any integration issues

Phase 4: Deployment
Duration: Week 4 (Mon–Wed)
Milestone: Fully functional, tested, and documented application
Deliverables: Final MVP app, complete documentation, testing suite, and code package
Week 4: Monday (DevOps, Frontend Dev)

- Final Testing
    - Write and run full unit test suite
    - Perform UI/UX testing for input/output flow
- CI Setup
    - Set up basic CI/CD pipeline (GitHub Actions or similar)

Week 4: Tuesday (DevOps, Frontend Dev)

- Documentation
    - Generate developer and user documentation using pdoc or MkDocs
    - Document API endpoints, model usage, setup instructions
- UI Finalization
    - Polish frontend UI for usability and clarity
    - Ensure error handling and edge case coverage

Week 4: Wednesday (All)

- Project Wrap-up
    - Conduct full end-to-end tests

- Export trained model(s) and save artifacts
- Package app and codebase using Poetry
- Final review and submission

VI. Reference

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